

Experiment 18 — Age & Body Fat

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In [1]: import pandas as pd
import matplotlib.pyplot as plt
import scipy.stats as stats

df = pd.DataFrame({
    "Age": [23, 23, 27, 27, 39, 41, 47, 49, 50, 52, 54, 54, 56, 57, 58, 58, 60, 61],
    "%Fat": [9.5, 26.5, 7.8, 17.8, 31.4, 25.9, 27.4, 27.2, 31.2,
34.6, 42.5, 28.8, 33.4, 30.2, 34.1, 32.9, 41.2, 35.7]
})

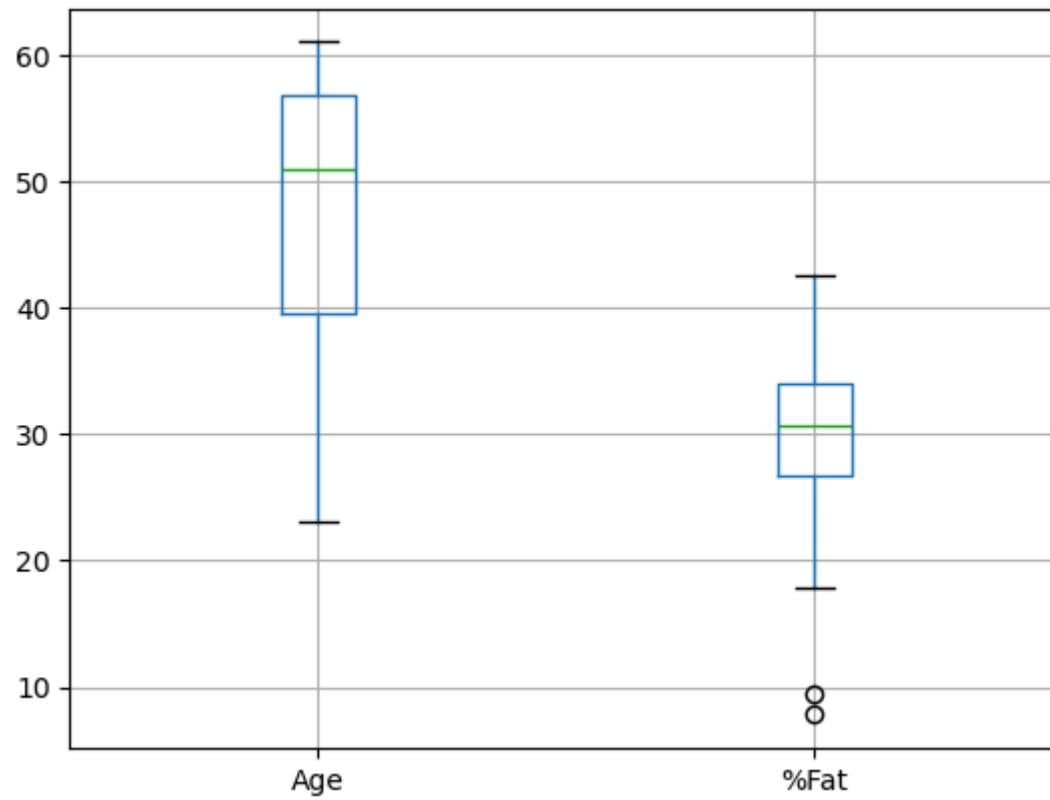
print(df.mean())
print(df.median())
print(df.std())

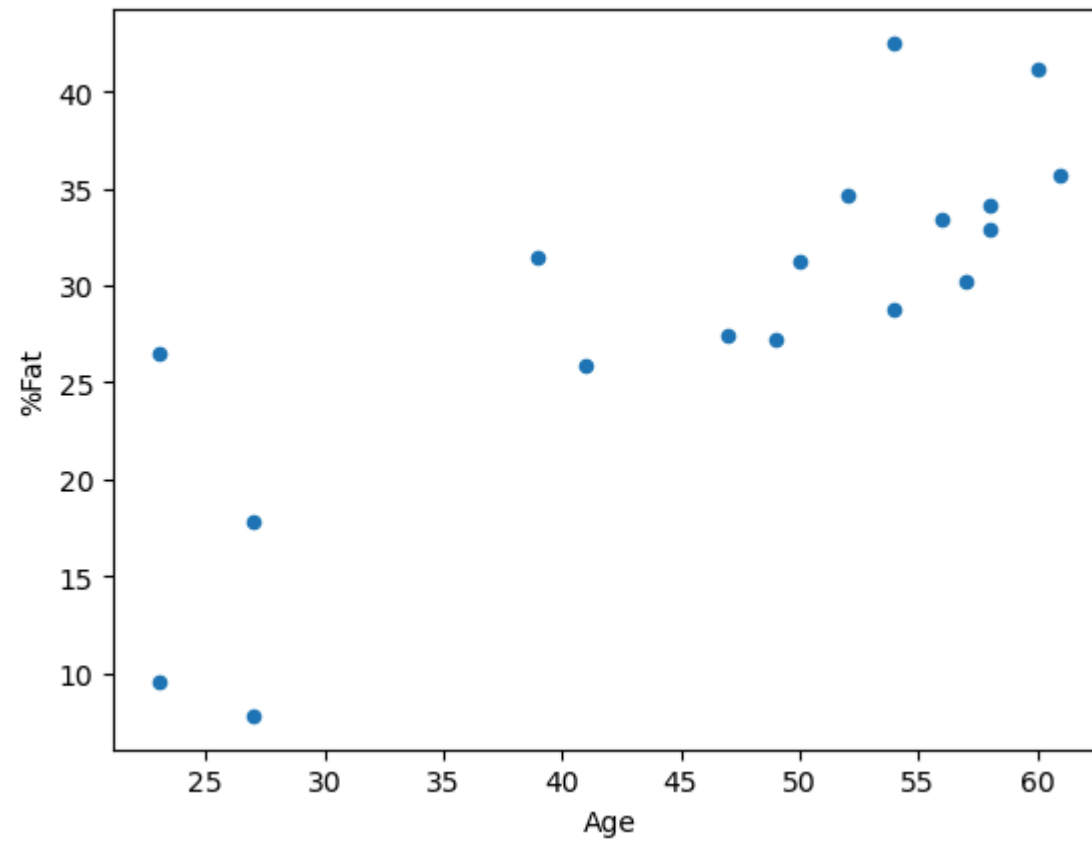
df.boxplot()
plt.show()

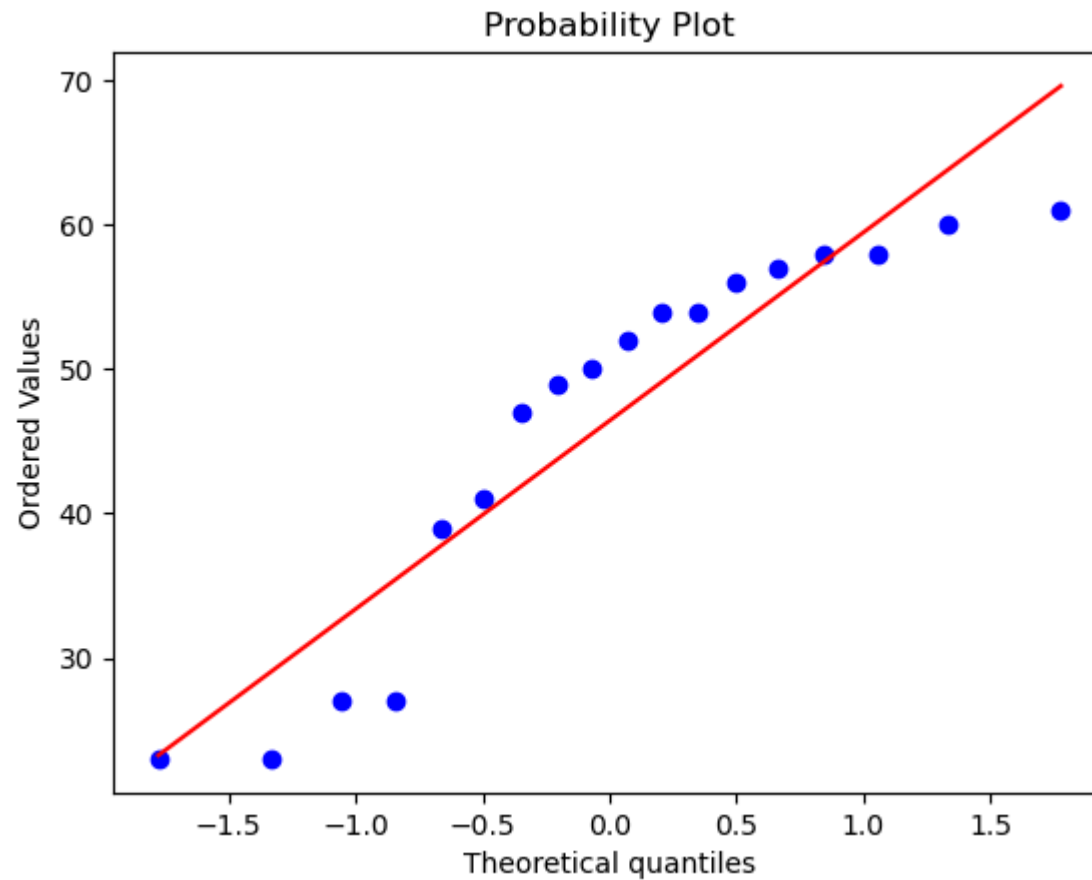
df.plot.scatter(x="Age", y="%Fat")
plt.show()

stats.probplot(df["Age"], dist="norm", plot=plt)
plt.show()
```

```
Age      46.444444
%Fat     28.783333
dtype: float64
Age      51.0
%Fat     30.7
dtype: float64
Age      13.218624
%Fat      9.254395
dtype: float64
```







EXPERIMENT 19 — SALES AND PROFIT ANALYSIS

```
In [4]: import pandas as pd

df = pd.read_csv("C:\\Users\\btaru\\Downloads\\sales_data.csv")

df["Total Sales"] = df["Quantity Sold"] * df["Unit Price"]

sales = df.groupby("Product")["Total Sales"].sum()
profit = sales * 0.20

print("Total Sales:\n", sales)
```

```
print("Overall Profit:", profit.sum())
print("Top 5:\n", profit.sort_values(ascending=False).head(5))
```

Total Sales:

Product

Headphones 25000

Keyboard 18000

Laptop 275000

Monitor 60000

Mouse 24000

Name: Total Sales, dtype: int64

Overall Profit: 80400.0

Top 5:

Product

Laptop 55000.0

Monitor 12000.0

Headphones 5000.0

Mouse 4800.0

Keyboard 3600.0

Name: Total Sales, dtype: float64

EXPERIMENT 20 — CUSTOMER SEGMENTATION

```
In [6]: import pandas as pd

df = pd.read_csv("C:\\Users\\btaru\\Downloads\\customer_data.csv")

low = df["Total Spending"].quantile(.33)
high = df["Total Spending"].quantile(.67)

df["Segment"] = df["Total Spending"].apply(
    lambda x: "High" if x >= high else
              "Medium" if x >= low else "Low"
)

print(df)
print(df.groupby("Segment")["Age"].mean())
```

	Customer ID	Age	Gender	Total Spending	Segment
0	101	22	M	1200	Low
1	102	35	F	5500	Medium
2	103	28	M	2300	Low
3	104	42	F	8200	High
4	105	31	F	4500	Medium
5	106	25	M	1800	Low
6	107	45	F	9500	High
7	108	38	M	6200	Medium
8	109	29	F	2800	Low
9	110	52	M	11000	High
10	111	34	M	5200	Medium
11	112	41	F	7600	High
12	113	26	M	1500	Low
13	114	48	F	8800	High
14	115	30	F	3900	Medium

Segment

High 45.6

Low 26.0

Medium 33.6

Name: Age, dtype: float64

In []: